

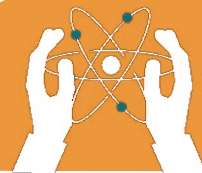
UNIT 6

SOLUTIONS, SUSPENSIONS, AND COLLOIDS

Learning objectives:

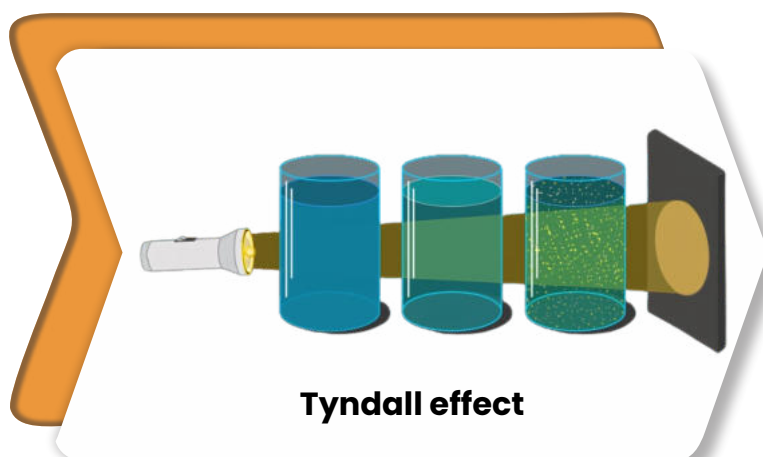
- Distinguish between solutions, suspensions, and colloids by properties.
- Explain the Tyndall effect in identifying mixtures.
- Identify examples of solutions, suspensions, and colloids.

SOLUTIONS, SUSPENSIONS, AND COLLOIDS



Have you ever mixed sugar in water? What happens?

Show three jars: one with saltwater (solution), one with flour



In everyday life, We encounter various mixtures daily, some uniform like saltwater, others that separate like muddy water. This chapter introduces three types of mixtures: **solutions, suspensions, and colloids**. It explains their unique properties, particle behaviour, and clarity differences. Additionally, it discusses the **Tyndall effect**, which helps distinguish between these mixtures

Key Concepts

Solution: A homogeneous mixture with dissolved particles.

Suspension: A heterogeneous mixture with visible particles that settle over time.

Colloid: A mixture with particles larger than in solutions but smaller than in suspensions, remaining evenly distributed.



SOLUTIONS, SUSPENSIONS, AND COLLOIDS

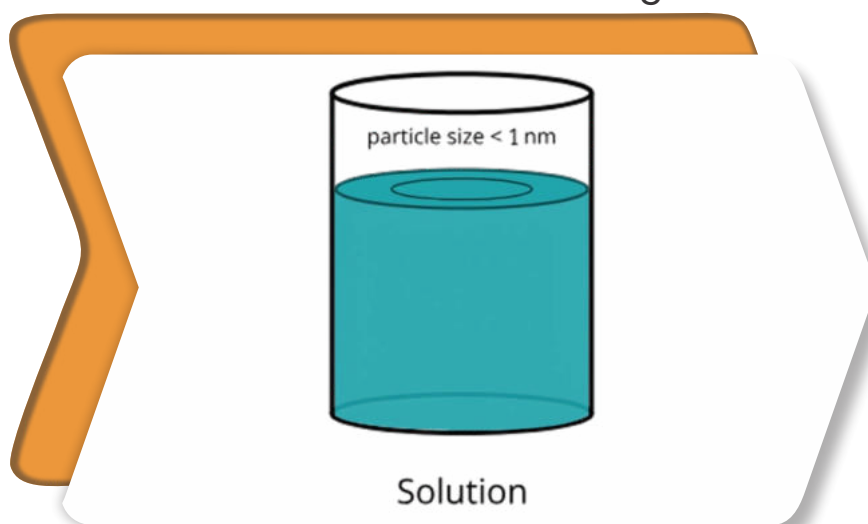
Solutions

A solution is a homogeneous mixture where one substance (solute) is completely dissolved in another (solvent). Solutions are clear and transparent.

Example: Saltwater, sugar dissolved in water, vinegar.

Properties of Solutions

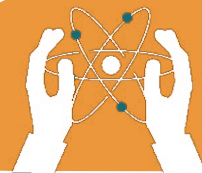
- **Particle Size:** Very small, typically less than 1 nanometre.
- **Visibility:** Particles cannot be seen with the naked eye.
- **Settling Behaviour:** Particles do not settle.
- **Tyndall Effect:** Solutions do not scatter light in this effect.



• DID YOU KNOW?

The Tyndall Effect is light scattering caused by particles in colloids and suspensions, making the beam visible. It does not occur in solutions, where particles are too small to scatter light. This effect helps distinguish colloids and suspensions from solutions. examples, like a flashlight through fog vs. clear water.





Activity I: Observing a Solution

Dissolve a teaspoon of salt in a glass of water. Observe if the salt is visible after stirring. Shine a flashlight through the glass to observe any light scattering.

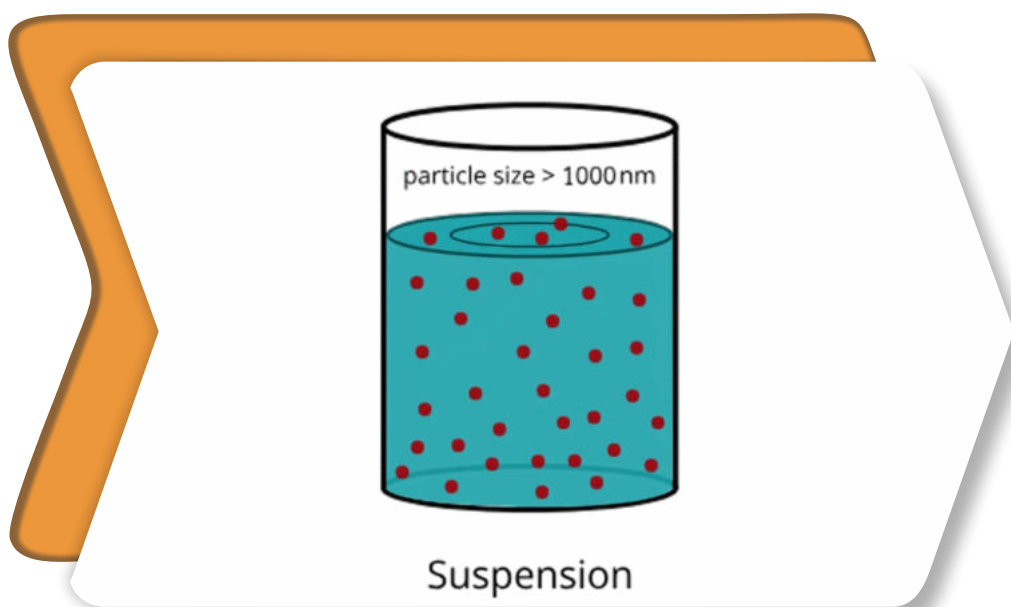
Suspensions

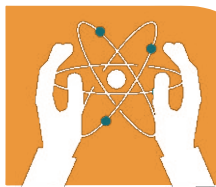
A suspension is a heterogeneous mixture where particles are large enough to be visible and may settle over time.

Example: Muddy water, flour in water, orange juice with pulp.

Properties of Suspensions

- **Particle Size:** Larger, typically more than 1000 nanometres.
- **Visibility:** Particles can be seen with the naked eye.
- **Settling Behaviour:** Particles settle upon standing.
- **Tyndall Effect:** Suspensions scatter light.





SOLUTIONS, SUSPENSIONS, AND COLLOIDS

Activity 2: Observing a Suspension

Mix a spoonful of flour in a glass of water and stir. Observe the water immediately after stirring and after a few minutes. Shine a flashlight through the mixture to see if the light is scattered.

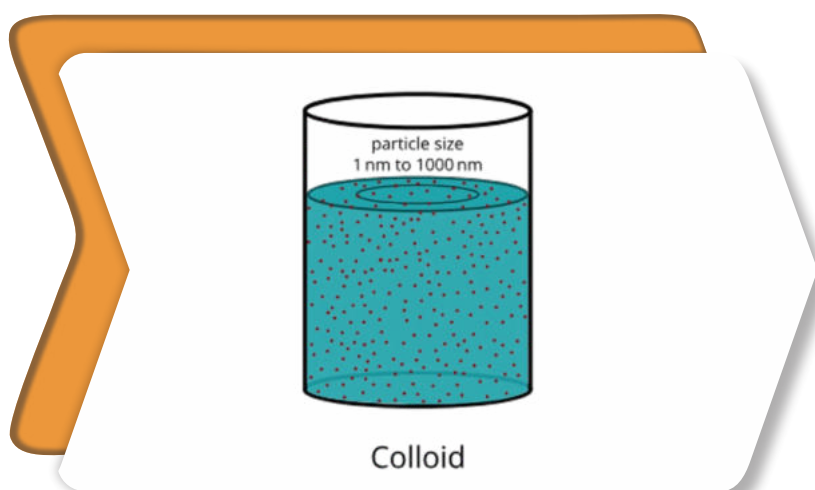
Colloid:

A colloid is a mixture where the particle size is intermediate between solutions and suspensions. The particles do not settle, and the mixture appears homogeneous but exhibits the Tyndall effect.

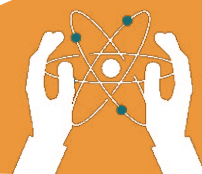
Example: Milk, mayonnaise, fog, shaving cream.

Properties of Colloids

- **Particle Size:** Intermediate, between 1 and 1000 nanometres.
- **Visibility:** Particles are not visible to the naked eye but can be seen under a microscope.
- **Settling Behaviour:** Particles do not settle on standing.
- **Tyndall Effect:** Colloids scatter light.



SOLUTIONS, SUSPENSIONS, AND COLLOIDS



Activity 3: Observing a Colloid

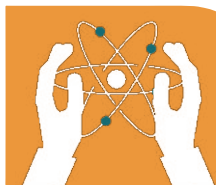
Take a glass of milk.

Shine a flashlight through the glass to observe light scattering (Tyndall effect).

Let the milk sit to see if particles settle over time.

Comparison Table: Solutions, Suspensions, and Colloids

Property	Solution	Suspension	Colloid
Appearance	Clear and transparent	Cloudy and opaque	Cloudy but uniform
Particle Size	Very small (<1 nm)	Large (>1000 nm)	Intermediate (1–1000 nm)
Settling Behaviour	Particles do not settle	Particles settle	Particles do not settle
Tyndall Effect	Does not scatter light	Scatters light	Scatters light

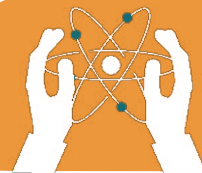


SOLUTIONS, SUSPENSIONS, AND COLLOIDS

Fill in the blanks.

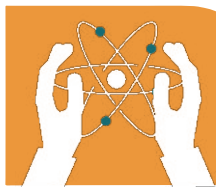
1. _____ is an example of a colloid where particles are intermediate in size and scatter light.
2. In a _____, the particles are large enough to be visible and tend to settle over time.
3. The _____ effect is the scattering of light by particles in a mixture, which occurs in colloids and suspensions.
4. _____ do not scatter light because their particles are too small to do so.
5. A _____ appears cloudy and uniform, with particles that do not settle but can scatter light due to the Tyndall effect.

SOLUTIONS, SUSPENSIONS, AND COLLOIDS



Complete the table below.

Property	Solution	Suspension	Colloid
Appearance		Cloudy and opaque	
Particle Size	Very small (<1 nm)		
Settling Behaviour			Particles do not settle
Tyndall Effect		Scatters light	



SOLUTIONS, SUSPENSIONS, AND COLLOIDS

Choose the correct answer.

1. Which of the following is a characteristic of a suspension?

- Particles are too small to settle
 - Particles do not scatter light
 - Particles are large and settle over time
 - Particles are dissolved completely
-

2. Which mixture type appears uniform but can scatter light due to small dispersed particles?

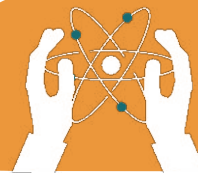
- Solution
 - Suspension
 - Colloid
 - Emulsion
-

3. What best describes a solution?

- Heterogeneous mixture with visible particles
 - Homogeneous mixture with dissolved particles
 - Cloudy mixture with particles that scatter light
 - Mixture where particles settle immediately
-

4. Which of the following will show the Tyndall effect?

- Saltwater
 - Air
 - Milk
 - Vinegar
-



Answer the Following Questions:

Short-Answer Questions

- Q1. Why don't particles settle in a colloid?
- Q2. Define any three properties of a suspension.
- Q3. What is the Tyndall effect?
- Q4. Does a solution scatter light? Why or why not?
- Q5. Give two examples of each type of mixture from daily life.

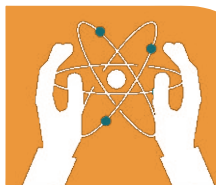
Long-Answer Questions

- Q1. Define solution, suspension, and colloid.
- Q2. Explain the Tyndall effect and identify which mixtures exhibit this effect. Why does salt dissolve completely in water, but flour does not?
- Q3. Describe one experiment to distinguish between a colloid and a suspension.



Critical Thinking

Why do colloids scatter light while solutions do not, and how could this property help us identify different mixtures in everyday life?



SOLUTIONS, SUSPENSIONS, AND COLLOIDS

Evaluate each statement as true or false. If true, explain why. If false, provide the correct reason.

Statement: "Solutions do not scatter light because their particles are too small."

Ans: _____

Statement: "A suspension will settle over time because its particles are large and visible."

Ans: _____

Statement: "Milk is a solution because it is uniform in appearance."

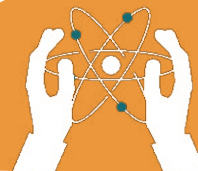
Ans: _____

Statement: "The Tyndall effect can be observed in colloids but not in solutions."

Ans: _____

Statement: "Saltwater is a suspension because salt particles are visible in the water."

Ans: _____



Activity Steps:

1. Solution Formation:

- Take a glass of water and dissolve a spoon of salt.
- Stir well and observe what happens.
- Record whether the particles settle down, remain mixed, or disappear.

2. Suspension Formation:

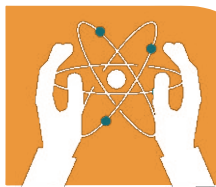
- Take another glass of water and mix in some chalk powder.
- Stir well and let it sit for a while.
- Observe what happens to the particles.

3. Colloid Formation:

- Take a glass of water and mix in some milk.
- Stir well and shine a flashlight through the glass.
- Observe whether the light scatters and if particles are visible.

Observation Table:

Mixture	Do particles settle down? (Yes/No)	Can you see the particles? (Yes/No)	Type of Mixture
Salt in water			
Chalk powder in water			
Milk in water			



SOLUTIONS, SUSPENSIONS, AND COLLOIDS

Complete the table with the properties of the types of mixture:

Property	Solution	Colloid	Suspension
Particle Size	very small, less than 1 nm		
Appearance			Opaque
Particle visibility		particles can not be seen with naked eye	
Homogeneity			heterogeneous
Sedimentation		does not settle on standing	
Permeability through filter paper	particles pass through filter paper		
Behaviour to light		scatters light, Tyndall effect	